Landform classification mapping using Remote sensing and GIS

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Abstract

The aim of this study is to evaluate the landform evolution over time in the region using a Geographic Information System (GIS) and remote sensing. The datasets for the required region were downloaded from earthexplorer platform which is the freely available resource. In this study the region was evaluated by applying different analytical GIS techniques including buffering ,layer stacking, mosaiqing ,clipping ,merging also the region was classified into different landforms like water body ,forest, barren land ,urban land and agricultural land. The study region includes the road connecting udupi to hassan via charmadi ghats through belur road .The accuracy assessment was done to check how accurate the land cover data were in terms of thematic categories .All the tools were performed using The Semi-Automatic Classification Plugin which is a free open source plugin in QGIS. After performing all the necessary tasks for four different years from 2000 to 2020 it was identified that there is no much change in the water body but the agricultural land has grown over time ,also the urban settlements are increased over time

Keywords

Landsat data, DEM data, Layer stacking, Mosaicing, clipping , landform classification, Accuracy assessment.

Introduction

A Landform is a three dimensional feature on the earth‘s surface formed by natural processes and are defined as ‘specific geomorphic features on the earth’s surface, ranging from large-scale features such as plains and mountain ranges to minor features such as sand and valleys. Natural phenomena and habitats of plants are directly related to landform patterns and the relative position with

a landform. comprehensive action of the internal and external forces have major influence on landform types.

Systematic acquisition of regional landform feature objects, analysis of landform feature parameters, and fine classification of landform are the research hotspots in geoscience research. Landform morphology exhibit significant difference reflection in different landform features, and accuracy analysis of landform features are key steps to describe the landform classification and to understand internal geomorphologic formation process in a specific region. In particular, correctly reflecting the macroscopic landform features is the basis for the accuracy of landform classification.

Early studies on landform classification, combined with the landform information, classification principle, and systems, can be traced back to manual and field observation. However, supervised (Prima et al., 2006) and unsupervised classification methods (Piloyan and Konečný, 2017) plays an import role in landform research to classify landforms on multiband images that composed of terrain elements. Based on the diversity of theoretical research, a variety of landform classification methods are found. Some methods are based on the morphological index (Arrell et al., 2007), that is, threshold classification of terrain factors, such as decision tree method (Iwahashi and Pike, 2007). Other techniques include methods which use digital elevation model (DEM) as the vital data source for describing terrain relief (Chen et al., 2014a; Tang et al., 2015) and landform spatial patterns (Chen, 2020).

Over the past few decades, experts and scholars all over the world have been focusing on remote sensing data classification methods, from supervised and unsupervised classifications based on traditional statistical analysis to widely used support vector machine , neural network and expert decision system. Among them, the pixel-based statistical classification technique has been dominant and relatively mature, and has achieved good application results in some fields. Supervised and unsupervised classifications are pixel-based classification, in which individual image pixels are analysed by the spectral information that they contain (Richards, 1993). This is the traditional approach to classification since the pixel is the fundamental (spatial) unit of a satellite image, and consequently it comes naturally and is often easy to implement. Various schemes are used in pixel-based classification. Maximum-likelihood, Minimum-distance-to-mean, and Minimum-Mahalanobis-distance are three of these. One assumes that the classes have been statistically characterized, so, for instance, their means and variances are known.

Landsat images provide considerable information about landforms, but little systematic work has been carried out on the usefulness of Landsat scenes for landform classification, and their relative merits compared with conventional tools such as topographic maps and air photographs. The classification presented here is based on supervised and unsupervised classification methods that enables Landsat images to be used for systematic landform study.

Study area

The Study area of our project is located in Karnataka state, India which includes the road connecting udupi to hassan via charmadi ghats through belur road namely, National Highway 66 ,State Highway 70 and National Highway 73 where hills rise in the central area . It includes charmadi ghat where the comprehensive effects of internal and external geological processes are evident . Charmady ghat one of the points in Western ghats lies on National Highway 73 which has experienced many landslides in past. The roads in this ghat section are very narrow . Therefore the study of landform features gain importance in this region in order to suspect the possible hazards to ensure a safe journey in this route .

The study is based on periodic change of past 20 years since 2000 to 2020

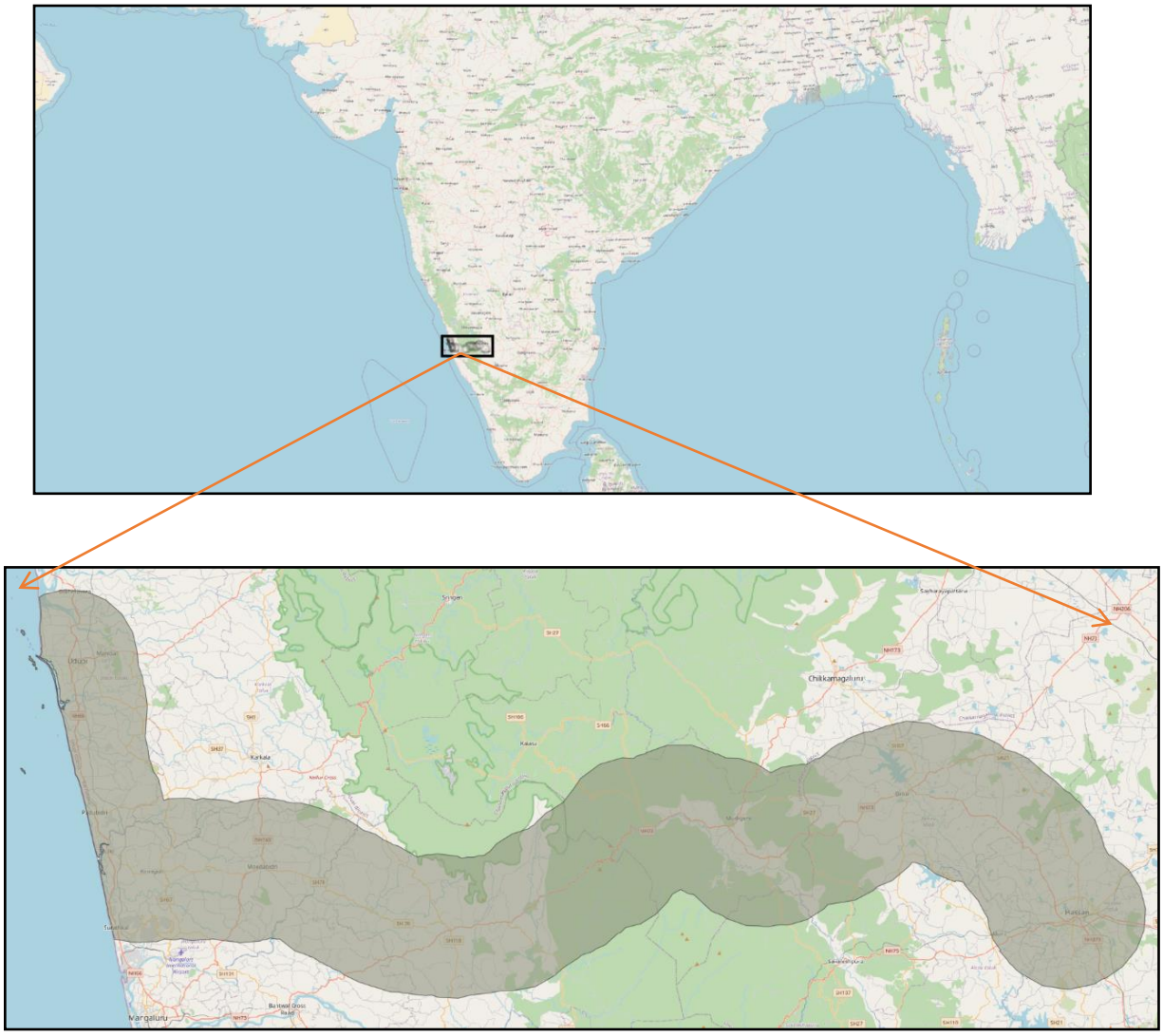


Fig . 1 Location of study area within India map

Fig. 2 Landslide event occurred on Aug 13 2019 in study area

Materials

Materials used in this project include,

- The Landsat images of different bands covering the study area were downloaded from USGS Earth Explorer platform .
- India shape file was downloaded .

We used the following steps of methodology

1. Collection of data and establishment of a spatial database

The primary GIS data capture techniques use remote sensing and surveying technologies to capture the data using either raster data capture or vector data capture. The raster GIS data capture technique involves capturing of attributes and other da physical contact. These captured raster and vector data are downloaded from USGS Earth explorer .

Landsat 7 Etm+ ,landsat 8 , landsta 4-5 images covering the study area of different years were used for processing .

2. The collected data were processed By various analytical GIS techniques, using Semi - Automatic Classification plugin in QGIS software

Semi- Automatic Classification plugin (SCP)

The **Semi-Automatic Classification Plugin** (SCP)was developed by Developed by Luca Congedo. It is a free open source plugin in QGIS that allows for the supervised and unsupervised classification of remote sensing images, providing tools for the download, the preprocessing and postprocessing of images.

The overall objective of SCP plugin is to provide a set of intertwined tools for raster processing in order to make an automatic workflow and ease the land cover classification, which could be performed also by people whose main field is not remote sensing.Search and download is available for ASTER, GOES, Landsat, MODIS, Sentinel-1, Sentinel-2, and Sentinel-3 images. Several algorithms are available for the land cover classification.

3. Buffer Zone of 15 km was created around the routes connecting udupi to hassan via charmadi ghats and study area shape file was created

Buffering is the most common operations in cartographic modeling. A buffer zone is an area that is within a given distance from a map feature. When you buffer on a set of features, the output is a set of polygons. These are usually Buffering points or lines that creates a new polygon layer .

Buffering usually creates two areas: one area that is within a specified distance to selected real world features and the other area that is beyond. The area that is within the specified distance is called the **buffer zone**. A buffer zone is any area that serves the purpose keeping real world features distant from one another.

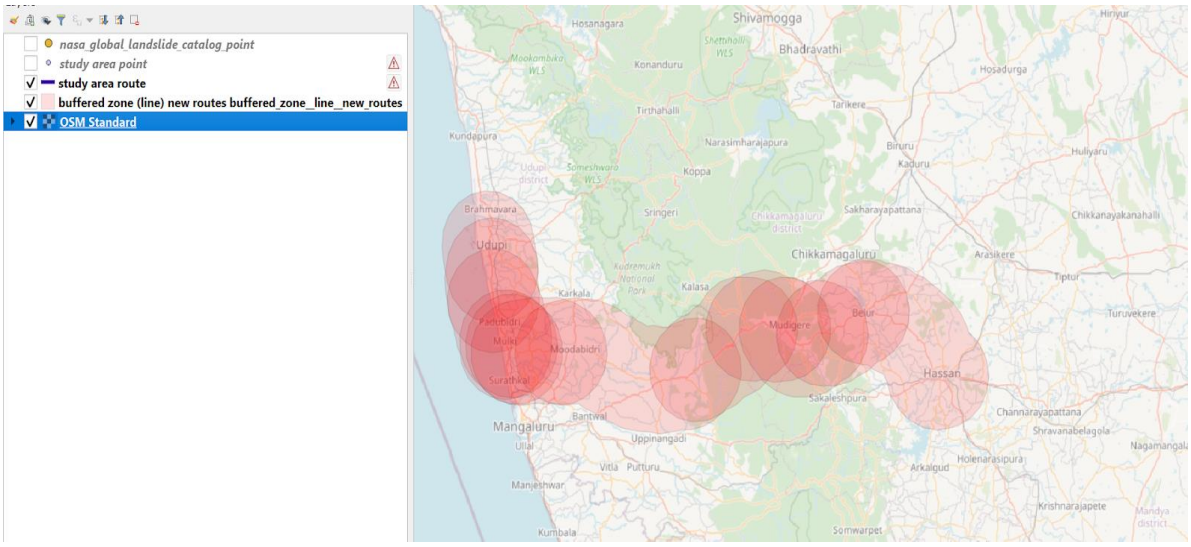


Fig. 3 Buffer zone of 15 km

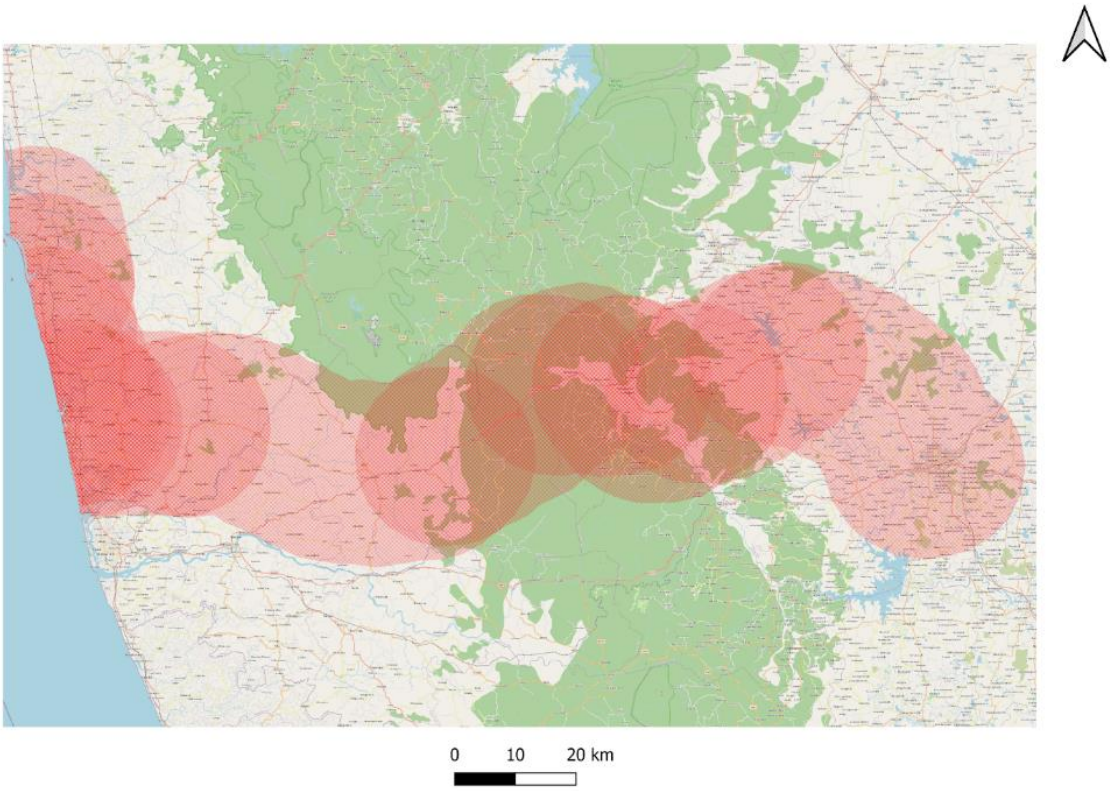


Fig 4 . Final Layot map of Shape file

4. Initially layer stacking was done to combine all the bands,then mosaiqing was performed to merge the two images in order to cover the study area,after that clipping was done to get our desired study area ,finally in order to get the single raster data set merging was performed.

Satellite Image Processing

Satellite image processing include various steps and processes to make image ready for interpretation and classification. These processes are important on each step of image processing such as geometric correction, radiometric correction, enhancement, classification etc. These processing methods are also depend on number of bands present in image as well as on spatial and temporal properties of satellite image.

In general satellite data need to stack, subset, mosaic and re-project based on the project requirement and data availability.

Layer stacking

Combining multiple separate bands in order to produce a new multi band image.

This type of multi band images are useful in visualizing and identifying the available Land Use Land Cover classes. In order to layer-stack, multiple image bands should have same extent.This process is primary process after acquiring satellite images as most of the satellite images are provided in different bands or layers, such as band-1 for blue wavelength, band-2 for green wavelength and so on.

If one need to make color composite than layer stacking tool is the process through which all the bands of that specific satellite image stacked in to a single file.

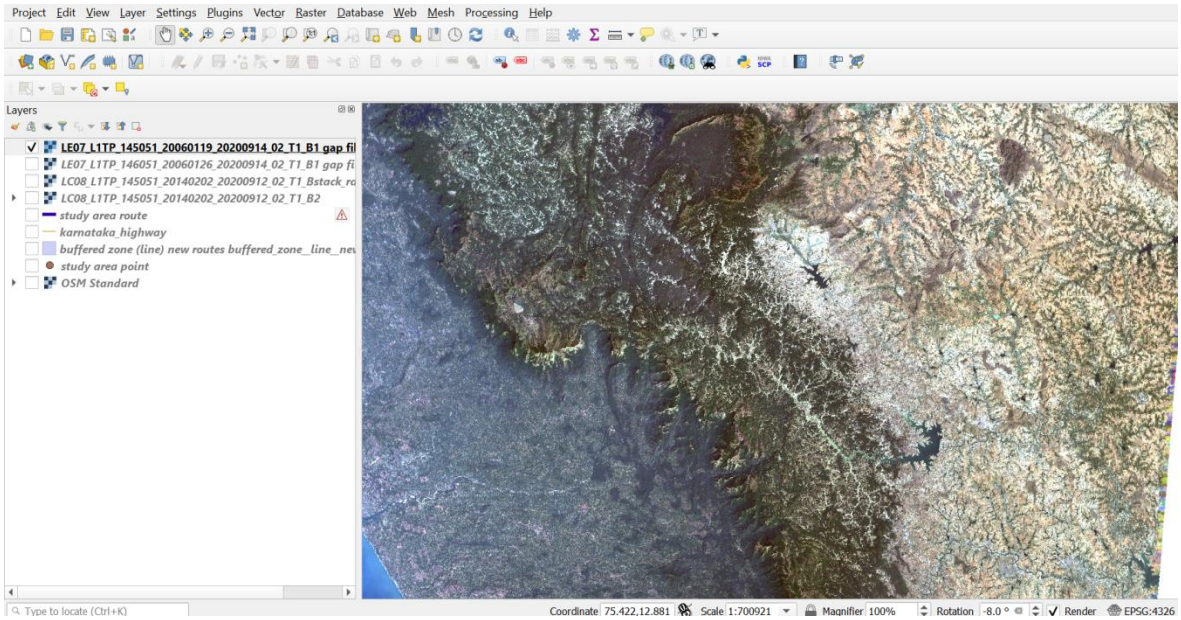


Fig . 5 Layer stacking of landsat 7 image (2006)

Fig .6 Landsat 7 Natural color image (Band combination – 3,2,1)

Fig .7 Landsat 7 colour Infrared (Band combination – 4,3,2)

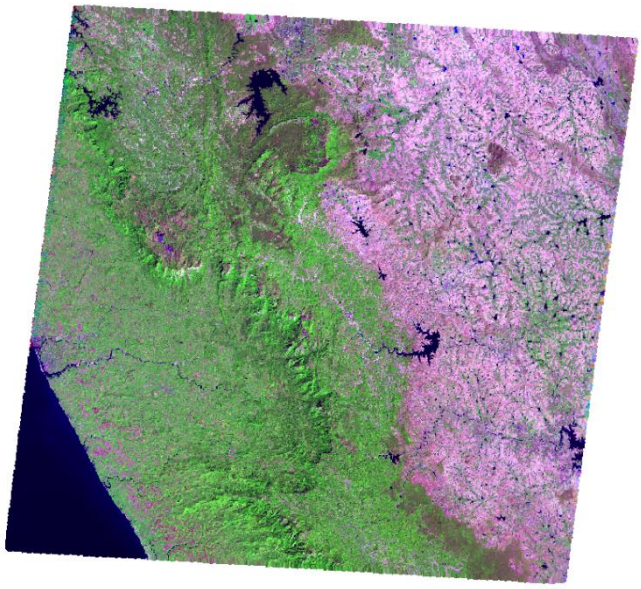


Fig .8 Landsat 7 Pseudo color image (Band combination – 5,4,3)

Band 1 (0.45 - 0.52u m): provides increased penetration of water bodies and also capable of differentiating soil and rock surfaces from vegetation and for detecting cultural features.

Band 2 (0.52 - 0.60u m): it is sensitive to water turbidity differences; it highlighted the turbid water in the Barkley Lake. Because it covers the green reflectance peak from leaf surfaces, it has separated vegetation (forest, croplands with standing crops)

from soil. In this band barren lands urban areas and roads and highways have appeared as brighter (lighter) tone, but forest, vegetation, bare croplands, croplands with standing crops have appeared as dark (black) tone. Also the Kentucky Lake has appeared as black tone.

Band 3 (0.63 - 0.69u m): senses in a strong chlorophyll absorption region and strong reflectance region for most soils. It has discriminated vegetation and soil. But it couldn't separated water and forest. Forest land and water both have appeared as dark tone. This band has highlighted barren lands, urban areas, street pattern in the urban area and highways

Band 4 (0.76 - 0.90u m): operates in the best spectral region to distinguish vegetation varieties and conditions. Because water is a strong absorber of near IR, this band has delineated water bodies (lakes and sinkholes), distinguished between dry and moist soils (barren land and croplands). In this band croplands and grasslands have showed higher reflectance (brighter tone) than the forest. Band 4 is useful for crop identification and emphasizes soil-crop and land-water contrast.

Band 5 (1.55 - 1.75u m): is sensitive to the turgidity or amount of water in plants. Band 5 has separated forest lands, croplands, water body distinctly. Forests have appeared as comparatively darker tone than the croplands (light gray).Band 5 has separated water body (dark tone) from barren lands, croplands, and grass lands (lighter tone). Since urban area and croplands have responded almost in same spectral reflectance band 5 could not be able to separate these areas.

Band 7 (2.08 -2.35u m): has separated land and water sharply. Band 7 has strong water absorption region and strong reflectance region for soil and rock. Urban area, croplands, highways, bare croplands have appeared as bright tone and water body, forest have appeared as dark tone.

Mosaicing

A **mosaic** is a combination or merge of two or more images. Image mosaic is normally done when a source map is larger than the scanner can handle. In this case, the map is scanned into sub-sections and then merged into a whole image for raster to vector conversion.

There are three groups of properties for a mosaic dataset: General, Default, and Key metadata. The general properties are similar to those you find for all raster datasets, such as the data source, extent, cell sizes, and bit depth. The default properties are specific to the mosaic dataset and include all the nonraster dataset-specific information. These properties affect how the mosaicked image will be presented to the user (or client) and how they might interact with it. The key metadata properties

are derived from the product definition, if one is defined for the mosaic dataset. It contains band and wavelength information used for rendering and some processing.

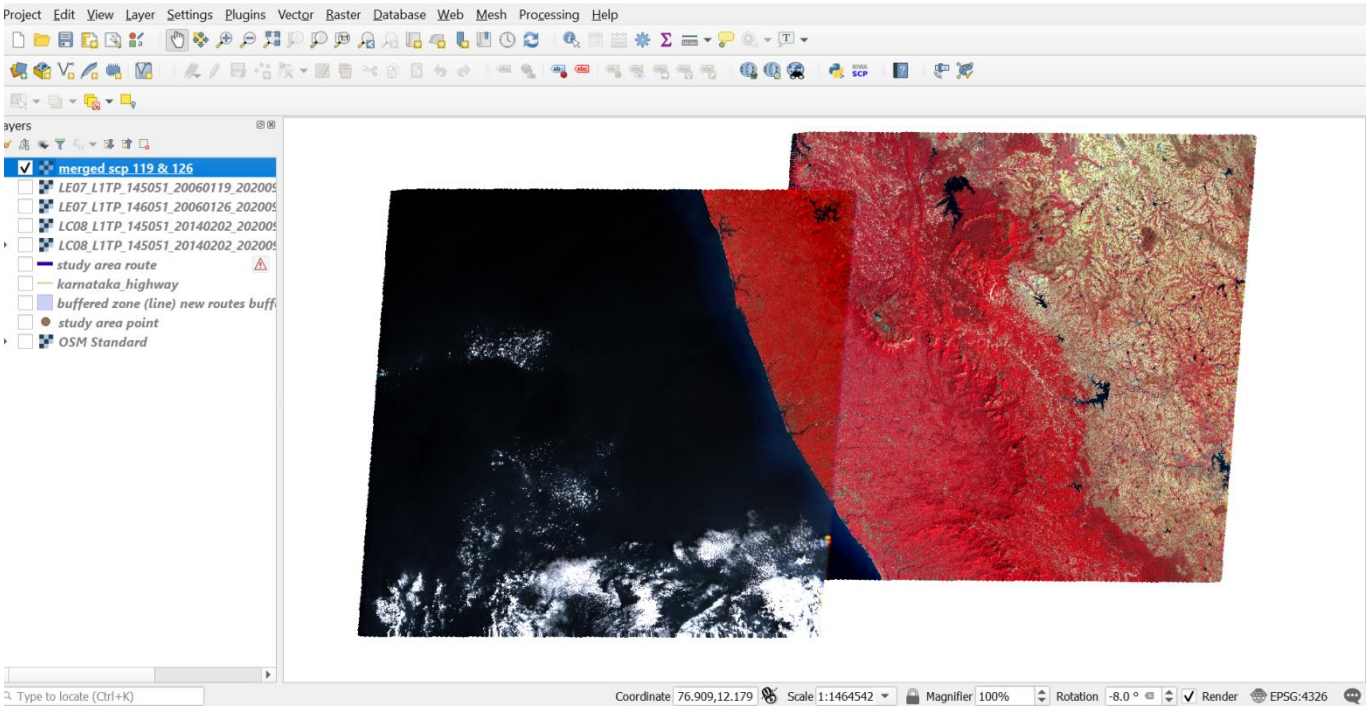


Fig . 9 Mosaic of landsat-7 2006 images

Clipping

Clipping is to overlay a polygon on one or more target features (layers) and extract from the target feature (or features) only the target feature data that lies within the area outlined by the clip polygon.

This tool allows us to extract a portion of a raster dataset based on a template extent. The clip output includes any pixels that intersect the template extent. To extract a portion of a feature dataset, use the Clip tool in the Analysis toolbox. An existing raster or vector layer can be used as the clip extent.

Initially study area shape file was used for clipping the mosaiced image and after that India shape file was used for further clipping in order to remove the sea part from the study area as sea part can create unnecessary problem during the classification of the image

Fig . 10 Clipping of landsat-7 2006 image

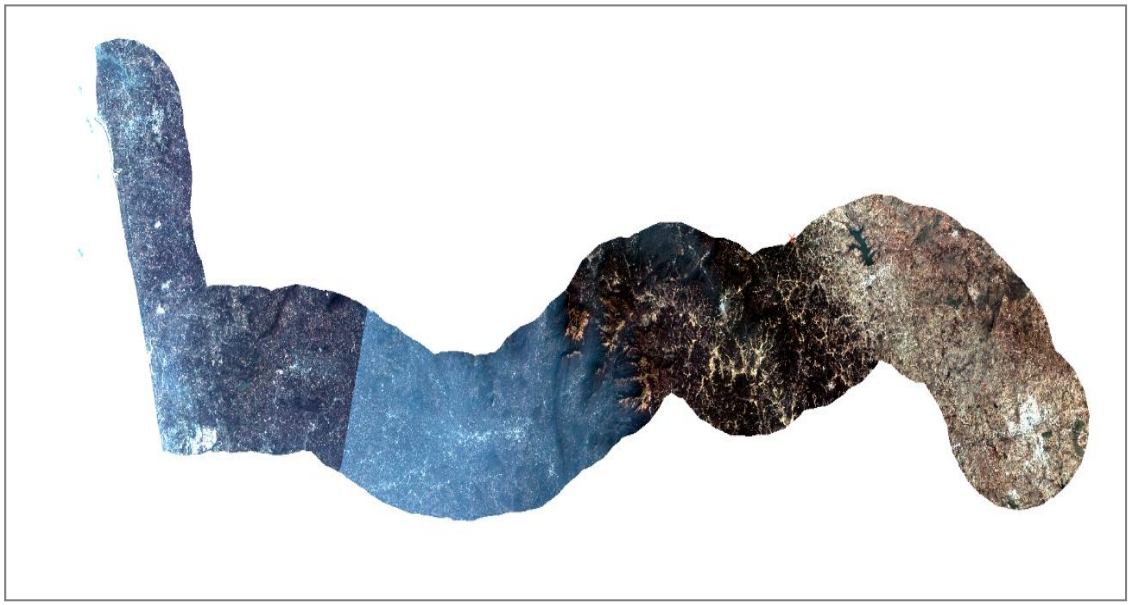


Fig . 11 Clipped image with sea part removed

5. using SCP plugin the obtained image was pan sharpened

Pansharpening

Pansharpening is a fusion technique to combine a panchromatic image of high spatial resolution with multispectral image data of lower spatial resolution to obtain a high-resolution multispectral image. By combining these two images, the final product not only has the higher resolution of the panchromatic dataset, but it also has the color associated with the multiband dataset, providing the user with better analysis opportunities.

During this process, the significant spectral characteristics of the multispectral data should be preserved. For images acquired at the same time by the same sensor, most algorithms for pansharpening provide very good results, i.e. they retain the high spatial resolution of the panchromatic image and the spectral information from the multispectral image (single-sensor, single-date fusion). For multi-date, multi-sensor fusion, however, these techniques can still create spatially enhanced data sets, but usually at the expense of the spectral consistency.

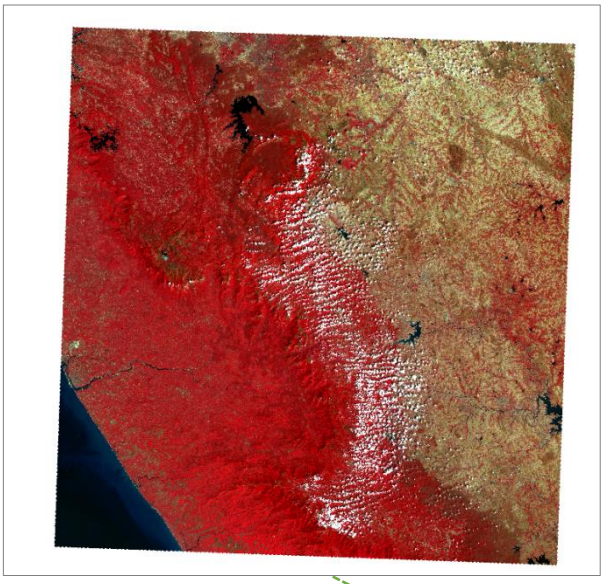


Fig. 12 Multispectral

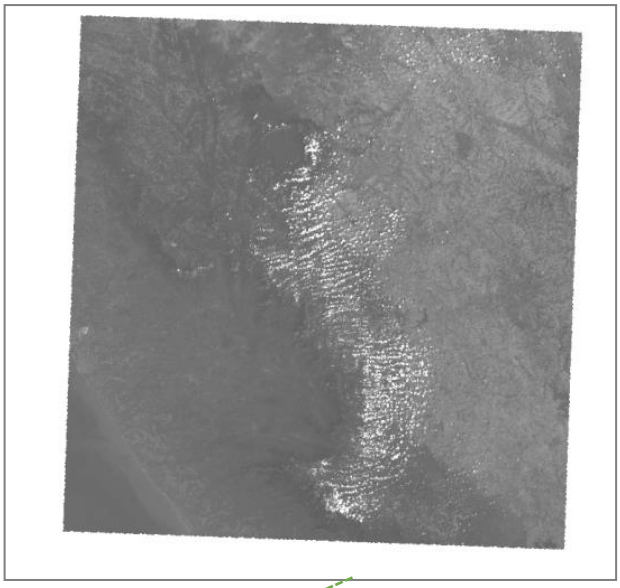


Fig. 13 Panchromatic

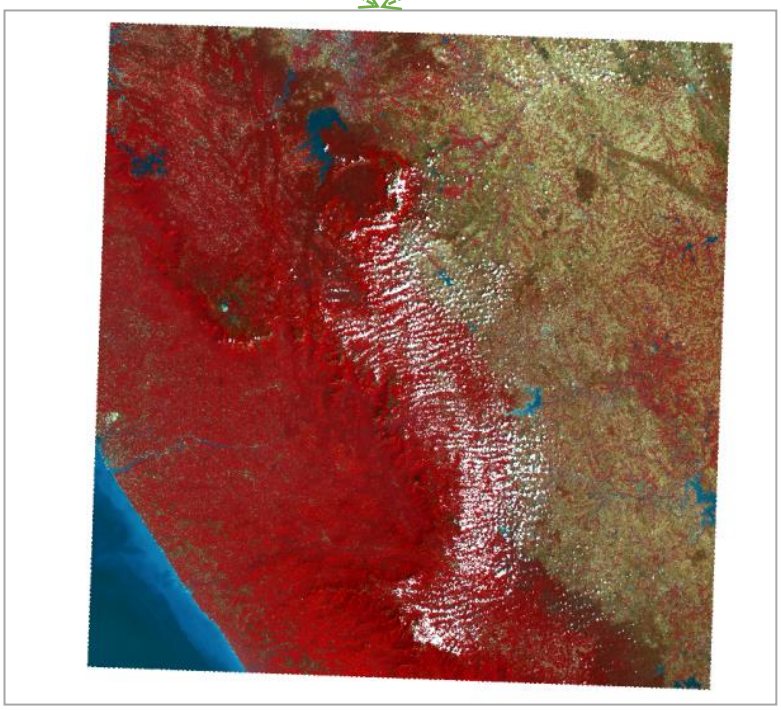


Fig. 15 Pansharpened image

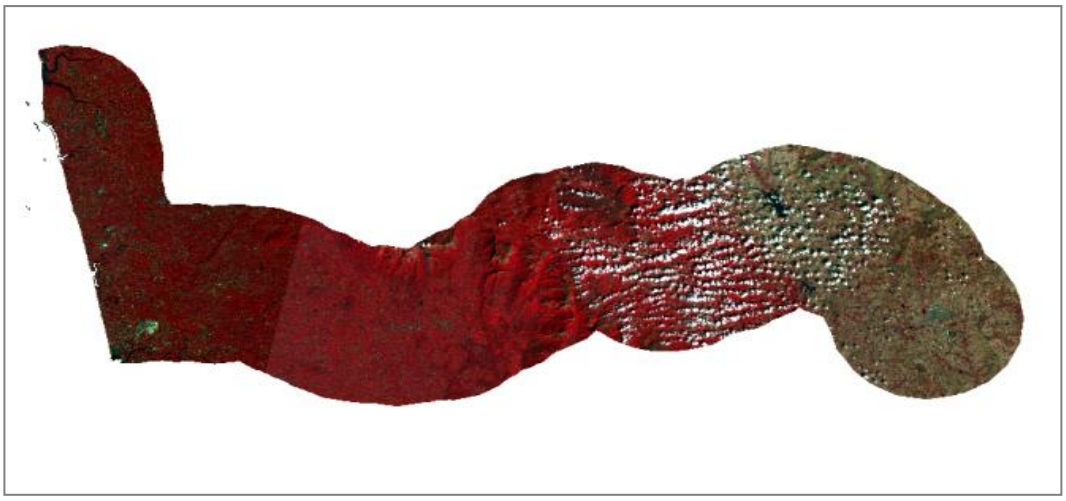


Fig 15 Landsat 8 (2014) Pansharpened image

6. Un supervised classification was performed using Semi Automatic Classification plugin using ISODATA method

Unsupervised classification

Unsupervised image classification is the process by which each image in a dataset is identified to be a member of one of the inherent categories present in the image collection without the use of labelled training samples. In this classification the outcomes (groupings of pixels with common characteristics) are based on the software analysis of an image without the user providing sample classes. The computer uses techniques to determine which pixels are related and groups them into classes. Here the desired number of output classes were specified and the final classified image was obtained.

Two common algorithms for creation of the clusters in unsupervised classification are k-means clustering and Iterative Self-Organizing Data Analysis Techinque (Algorithm), or ISODATA. We have used ISODATA algorithm to classify the image using Semi Automatic plugin in QGIS .

ISODATA unsupervised classification calculates class means evenly distributed in the data space then iteratively clusters the remaining pixels using minimum distance techniques. Each iteration recalculates means and reclassifies pixels with respect to the new means. Iterative class splitting, merging, and deleting is done based on input threshold parameters. All pixels are classified to the nearest class unless a standard deviation or distance threshold is specified, in which case some pixels may be unclassified if they do not meet the selected criteria. This process continues until the

number of pixels in each class changes by less than the selected pixel change threshold or the maximum number of iterations is reached.

Data was classified into 4 classes namely,
1 - Water Body, 2 – Urban, 3 - Thick Forest, 4 - Farmland/Agriculture

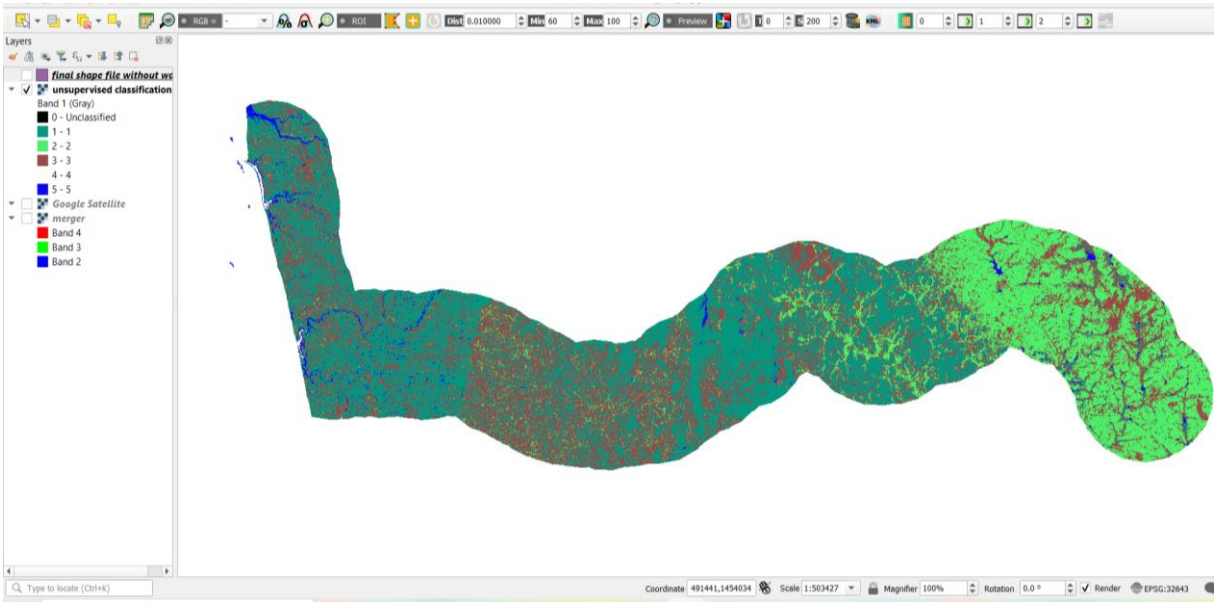


Fig 16 Unsupervised classification of images from 2001 to 2005

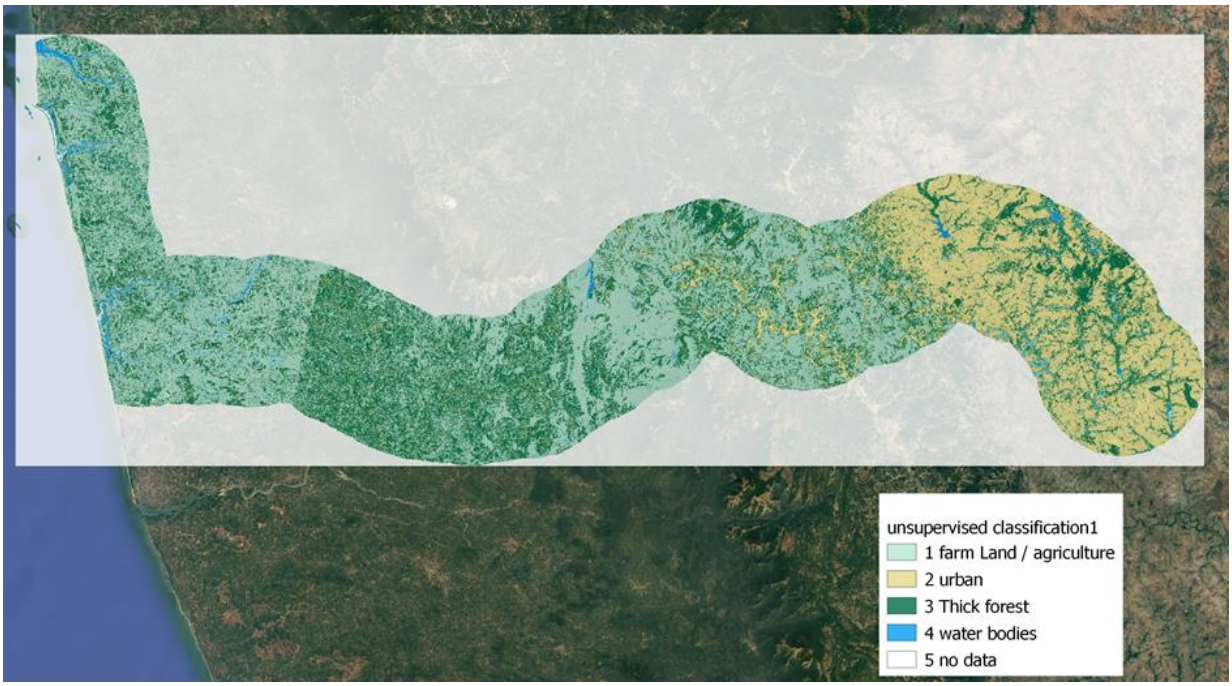


Fig 17 Unsupervised classification with legends (20016-2020)

7. Supervised classification was done using Semi Automatic Classification plugin in

QGIS software I which Samples were collected from each class

Supervised classification

Supervised classification is based on the idea that a user can select sample pixels in an image that are representative of specific classes and then direct the image processing software to use these training sites as references for the classification of all other pixels in the image.

The main aim of this classification is to automatically segregate pixels of a remote sensing image into groups of similar spectral character. The technique is used for the quantitative analysis of remote sensing image data. It is the concept of segmenting the spectral domain into regions that can be associated with the ground cover classes of interest to a particular application. In practice those regions may sometimes overlap.

This classification requires training data which are typical and homogeneous and the application of a set of methods, or decision rules. Training samples are key because they will determine which class each pixel inherits in your overall image. The most common supervised classification methods include: Maximum likelihood, Iso cluster, Class probability, Principal components and Support vector machine (SVM)

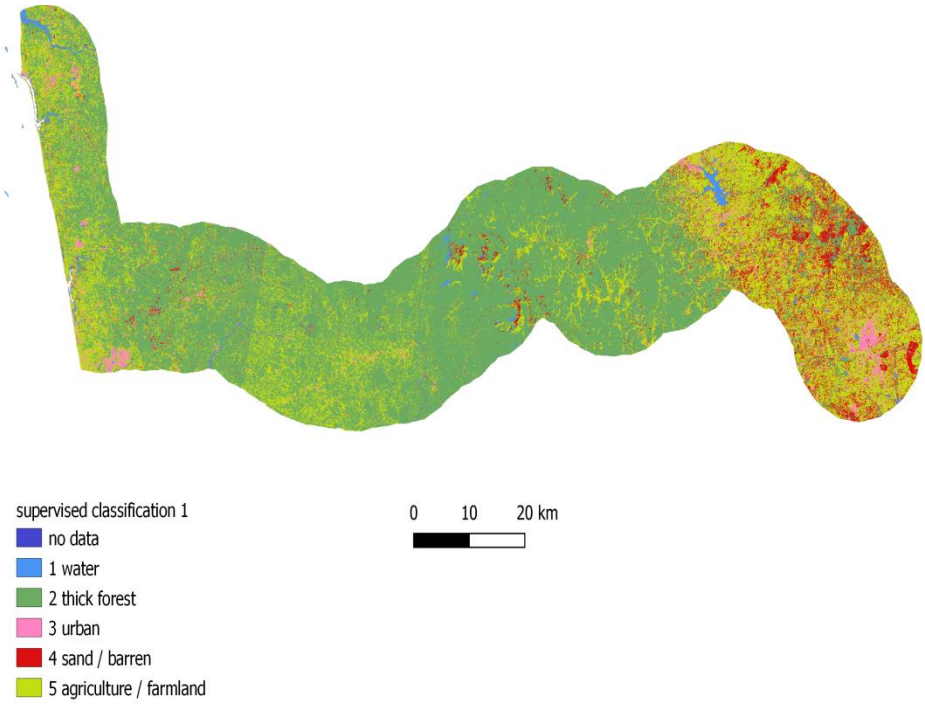


Fig .18 Supervised classification of data from 2001 to 2005

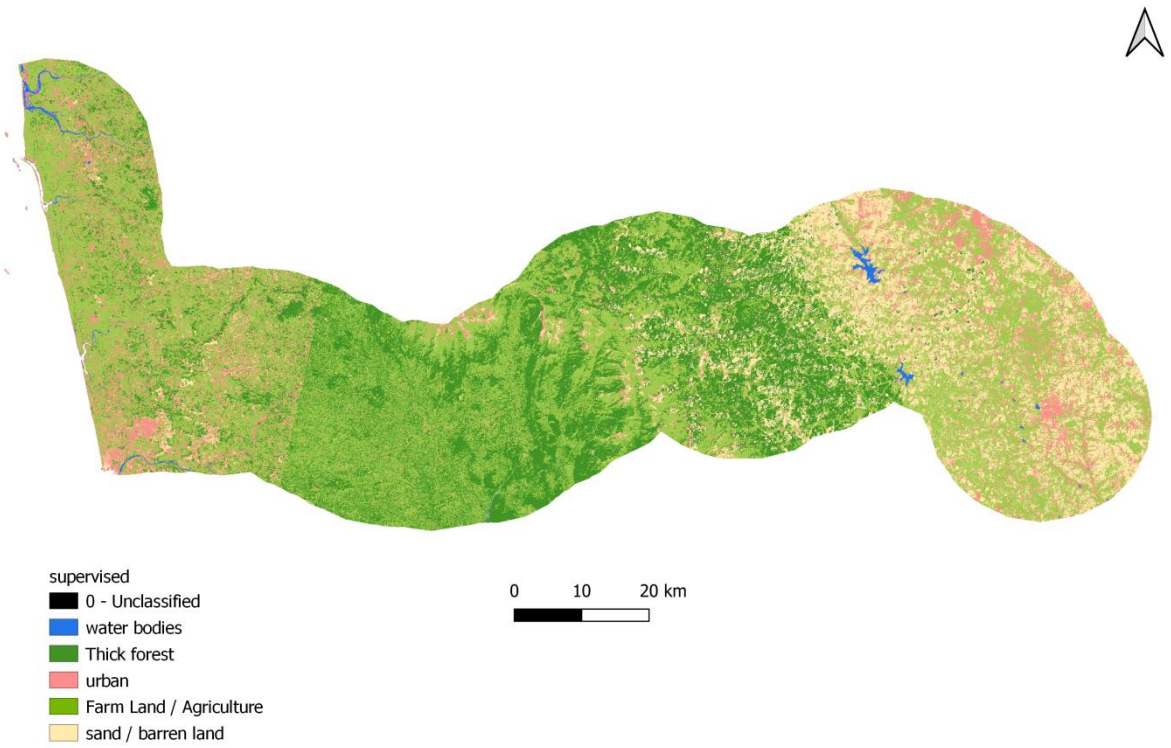


Fig .19 Supervised classification of data from 2006 to 2010

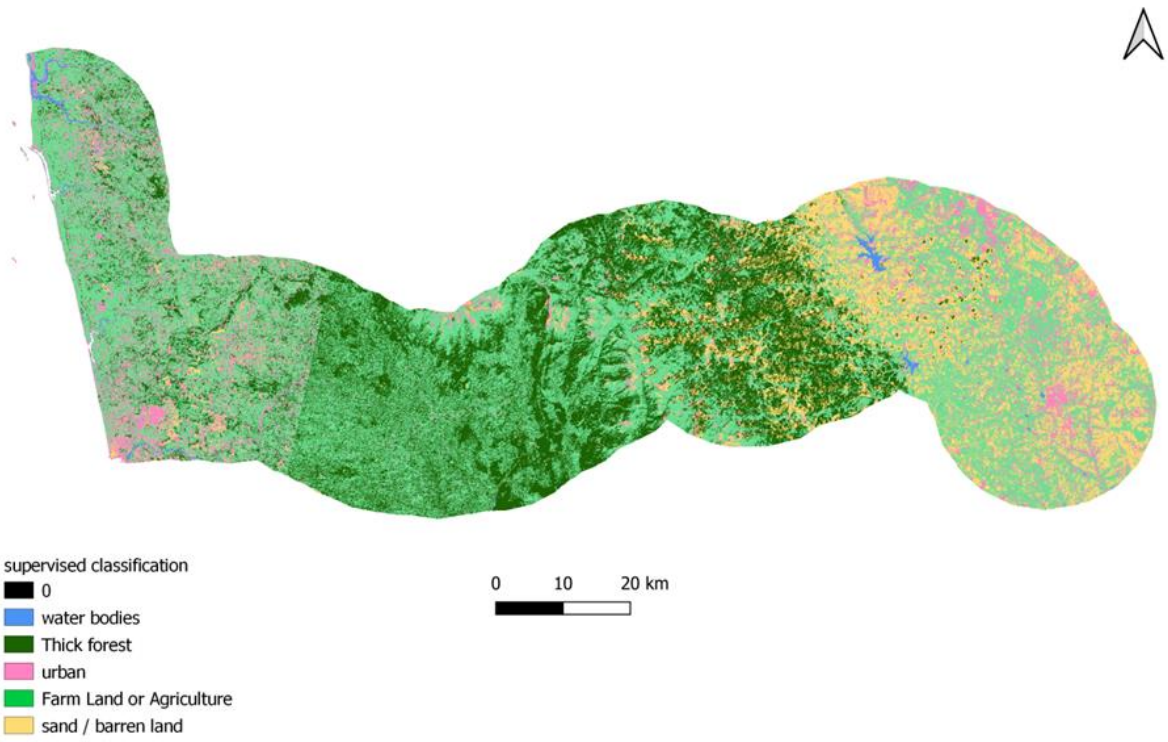


Fig .20 Supervised classification of data from 2011 to 2015

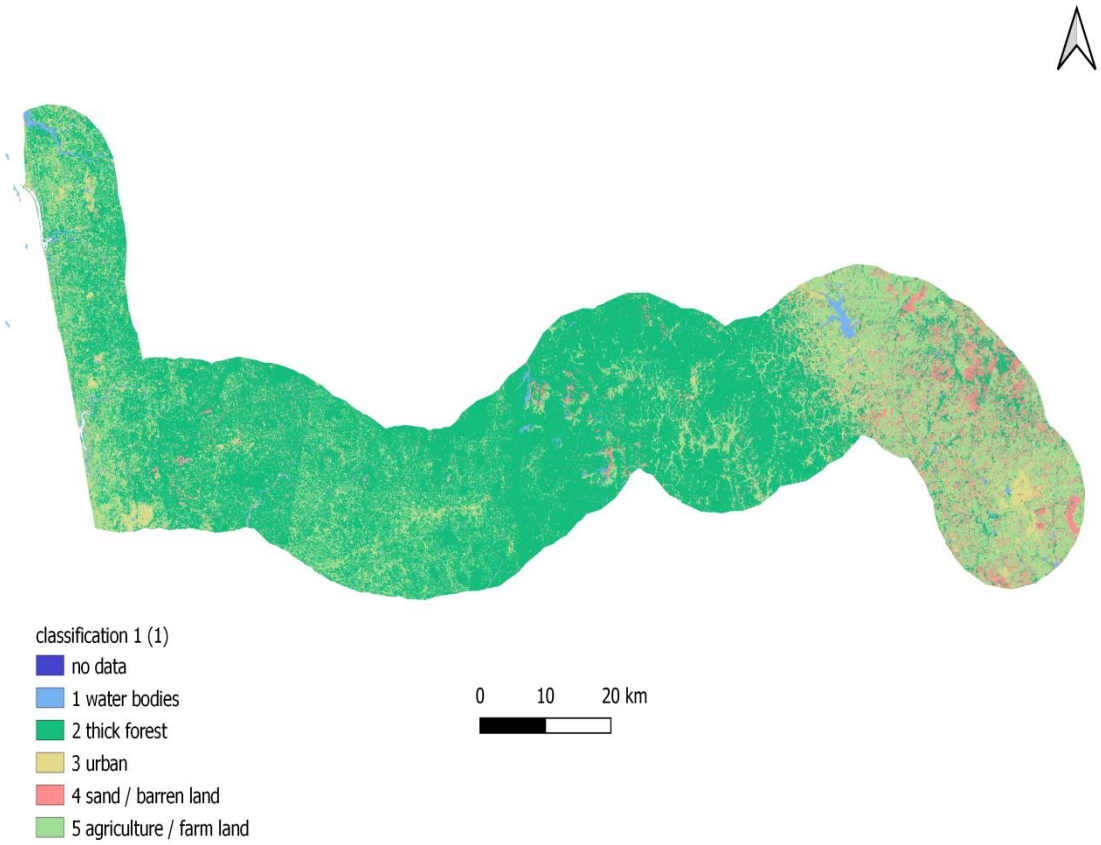


Fig .21 Supervised classification of data from 2016 to 2020

Classification report

2001 to 2005

Class	PixelSum	Percentage %	Area [metre^2]
0	657533	60.92170712495316	6022469700
1	60715	0.5527591617907812	54643500
2	2516353	22.909283950419464	2264717700
3	165937	1.5107172367631867	149343300
4	310954	2.8309754162149483	279858600
5	1238396	11.274557109858458	1114556400

2006 to 2010

Class	PixelSum	Percentage %	Area [metre^2]
0	27114076	51.191077514976	6100667100
1	118870	0.22442525366548347	26745750
2	6805820	12.849313366716755	1531309500
3	2963547	5.595144167784828	666798075
4	12205032	23.042966287501834	2746132200
6	3759065	7.097073409355099	845789625

2011 to 2015

Class	PixelSum	Percentage %	Area [metre^2]
0	23114076	54.191077514976	6200873170
1	98870	0.324425253665	23875750
2	6305820	11. 366716755	1371379732
3	2963547	6.587134167784341	684798075
4	10206482	20.67298728871876	2416132245
6	3859985	9.874033567997	894789155

2016 to 2020

Class	PixelSum	Percentage %	Area [metre^2]
0	6691633	60.92170712495316	6022469700
1	60715	0.5527591617907812	54643500
2	2516353	22.909283950419464	2264717700
3	165937	1.5107172367631867	149343300
4	310954	2.8309754162149483	279858600
5	1238396	11.274557109858458	1114556400

8. Finally accuracy assessment was performed in Semi Automatic Classification plugin for the supervised classification

Accuracy assessment

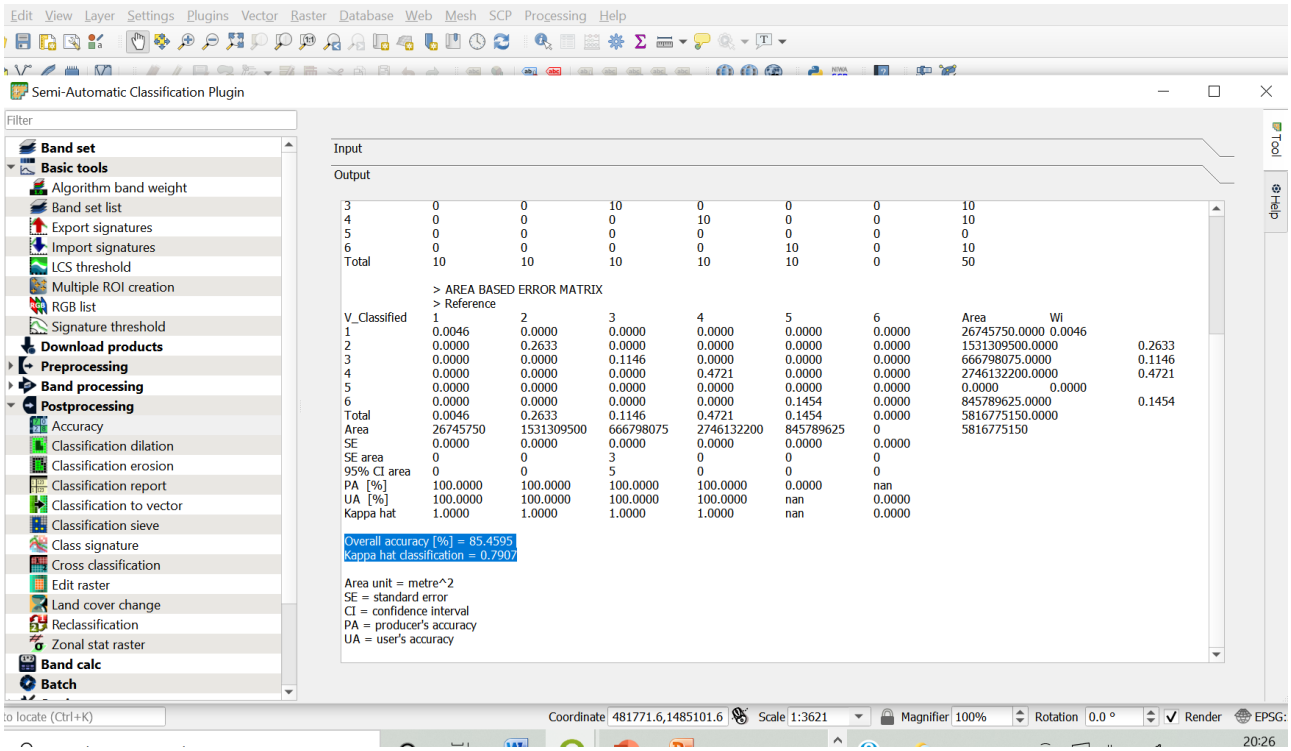
It is important to note that there are always errors in land cover maps derived from the remote sensing data. Accuracy assessment is the process of estimating how accurate the land cover data are in terms of thematic categories, in order to determine whether the product meets the requirements of the intended application. Accuracy assessment is used for comparing the land cover classification results in geospatial data that are assumed to be true. The assumed-true data are derived from ground referencing data or other reliable data sources.

The ***Kappa Statistic*** or Cohen's* Kappa is a statistical measure of inter-rater reliability for categorical variables. In fact, it's almost synonymous with inter-rater reliability. Kappa is used when two raters both apply a criterion based on a tool to assess whether or not some condition occurs. It is a measure of how the classification results compare to values assigned by chance. It can take values from 0 to 1.

If kappa coefficient equals to 1, then the classified image and the ground truth image are totally identical

Cohen suggested the Kappa result be interpreted as follows: values ≤ 0 as indicating no agreement and 0.01–0.20 as none to slight, 0.21–0.40 as fair, 0.41– 0.60 as moderate, 0.61–0.80 as substantial, and **0.81–1.00** as almost perfect agreement.

Accuracy of 2006-2010 classification



Similarly accuracy assessment of other classifications were conducted and the result obtained are as follows,

Year	Accuracy	kappa coefficient
2001-2005	65.6523	0.58
2006-2010	85.4595	0.79
2011-2015	75.8435	0.72
2016-2020	75.3225	0.69

Conclusion

The study area (Chosen road) has been classified into 5 classes ,water body, urban land, barren land ,forest area and agricultural land using SCP plugin .The western and central part of the region is covered by dense forest. The barren land is mainly concentrated in eastern part of the region due to acute rainfall. Eastern part of the area is mostly urban region.

Based on the accuracy assessment results we can conclude that ,

- Classification of 2001-2005 data has the kappa value of 0.58 which means the level of agreement is moderate
- Classification of 2006-2010 data has the kappa value of 0.79 approximately 0.8 which means the level of agreement is substantial
- Classification of 2011-2015 data has the kappa value of 0.72 which means the level of agreement is moderate
- Classification of 2016-2020 data has the kappa value of 0.69 which means the level of agreement is of moderate

Pixels in water bodies were less incorrectly classified as compared to other areas. Several pixels in sand/barren land were incorrectly classified to urban and vise-versa, resulting in mixing up of urban and sand region

The quantity of correct classified counts played a dominated position, causing the overall accuracy and kappa coefficien. Therefore, using Pixel based (supervised and unsupervised classification) is an effective method for improving landform classification accuracy

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